CHAPTER C

# Assessments and annual preparation

The T1 and T2 versions are available to students. The A and B versions and all the solutions are reserved for the teacher. The papers are original; they are not presented as official past papers. The approaches change between versions: direct or inverse model, optimisation or relative position, direct or reversed tree, projection or locus.

# EV1-T1 – Common test

**Practice paper**. Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

## Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 9 C. 8 D. 15
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 8 C. 2 D. 1
4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 60 D. 70
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 60 B. 64 C. 96 D. 100
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. $\frac{9}{10}$ D. 1
7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. 6 D. $-6$
8. The slope between $(0, 3)$ and $(2, 9)$ is
   A. 2 B. $-3$ C. 6 D. 3
9. The mean of $3, 5, 7$ is
   A. 6 B. 15 C. 3 D. 5
10. A sector of $90^\circ$ represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{3}{4}$ D. $\frac{1}{3}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$
12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 7 B. 5 C. 12 D. $-1$

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – A book collection (5 points)

A library has 120 books at the end of the reference month, called month 0. At the end of each following month, it adds 18 books to its stock; no book is removed. Let $u_n$ be the number of books available at the end of month $n$, for every natural number $n$.

1. Determine the type of $(u_n)$, its common difference and its expression in terms of $n$. (1 pt)
2. Calculate $u_8$, then $u_0 + u_1 + \cdots + u_8$. Explain what this sum accumulates and why it does not count books that are all distinct. (2 pts)
3. Determine the first month in which the stock strictly exceeds 300 books. Check the previous month. (2 pts)

### Exercise 2 – Area and choice of form (5 points)

A rectangle has a perimeter of 28 m. If one of its sides measures $x$ metres, the other measures $14 - x$ metres. Its area is modelled by the function $A$ defined on $[0, 14]$ by $A(x) = x(14 - x)$, in square metres. The endpoints represent degenerate rectangles.

1. Expand, then write A in vertex form. (1 pt)
2. Determine the maximum area and the corresponding value of x without a derivative. (2 pts)
3. Solve $A(x) \geq 40$ on the domain. (2 pts)

### Exercise 3 – A right angle (4 points)

The plane is equipped with an orthonormal frame. Given the points $A(0, 1)$, $B(3, 2)$ and $C(-1, 4)$. Lengths are expressed in the unit of the frame and areas in area units.

1. Calculate the coordinates of the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$. (1 pt)
2. Prove that these vectors are orthogonal and state the nature of the triangle. (1 pt)
3. Calculate the length $BC$ then the area of the triangle $ABC$. (2 pts)

# Solutions EV1-T1

**MCQ:** 1 B, 2 C, 3 A, 4 C, 5 D, 6 D, 7 B, 8 D, 9 D, 10 B, 11 B, 12 A. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

## Exercise 1 – A book collection

1. $u_n = 120 + 18n$, arithmetic with common difference 18. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $u_8 = 264$; nine terms, sum $\dfrac{9(120+264)}{2} = 1728$. This sum accumulates monthly states, not distinct books. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $120 + 18n > 300 \iff n > 10$. First index 11; $u_{10} = 300$, $u_{11} = 318$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 2 – Area and choice of form

1. $A = -x^2 + 14x = 49 - (x-7)^2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $A \leq 49$, equality at 7. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $49 - (x-7)^2 \geq 40 \iff (x-7)^2 \leq 9$, so $[4, 10]$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 3 – A right angle

1. $\overrightarrow{AB} = (3,1)$, $\overrightarrow{AC} = (-1,3)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Product $-3 + 3 = 0$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. $BC = \sqrt{(-4)^2 + 2^2} = 2\sqrt{5}$; perpendicular sides of norm $\sqrt{10}$, area 5. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

## EV1-T2 – Common test

**Practice paper**. Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

### Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $4 + 2 \times 3$.
   A. 18 B. 9 C. 13 D. 10
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^4}{2^2}$.
   A. 2 B. 4 C. 1 D. 9
4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 10 D. 60
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 100 B. 96 C. 64 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. 1 B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. $\frac{9}{10}$
7. The solution of $2x + 4 = 0$ is
   A. 2 B. 8 C. $-2$ D. $-8$
8. The slope between $(0, 4)$ and $(2, 10)$ is
   A. 3 B. $-3$ C. 6 D. 2
9. The mean of $4, 6, 8$ is
   A. 18 B. 6 C. 7 D. 4
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 4$, $f(-2)$ is
    A. 6 B. 16 C. 8 D. 0

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Two growth processes (5 points)

We compare two models for the output of a workshop. On day 0, each forecasts 30 items. Model A forecasts 10 extra items each day; model B forecasts a daily doubling. The outputs on day $n$ are therefore $a_n = 30 + 10n$ and $b_n = 30 \times 2^n$, for $n \in \mathbb{N}$. A running total here means the sum of the outputs of the days indicated.

1. State the type of each sequence and calculate the outputs on day 2. (1 pt)
2. Calculate the cumulative outputs for days 0 to 3 in each model. (2 pts)
3. Check that model B exceeds model A on day 1, then explain why this gap stays positive on all following days by comparing the daily increments. (2 pts)

### Exercise 2 – Imposed roots (5 points)

We look for a polynomial function $f$ of degree 2, defined on $\mathbb{R}$. Its curve crosses the x-axis at the x-coordinates $-1$ and 5, and it passes through the point $(0, 10)$. This information must be enough to determine the function.

1. Give its factorised form. (1 pt)
2. Determine its sign then solve $f(x) \leq 0$. (2 pts)
3. Give its vertex form and its maximum. (2 pts)

### Exercise 3 – A controlled approximation (4 points)

The function $f$ is defined on $\mathbb{R}$ by $f(x) = x^2 + 1$. We wish to approximate $f(3.02)$ using the tangent to its curve at the point with x-coordinate 3. The calculations are done without a calculator.

1. For $h \neq 0$, calculate and simplify the rate of change between 3 and $3 + h$. (1 pt)
2. Deduce $f'(3)$, then the equation of the tangent at the point with x-coordinate 3. (1 pt)
3. Use the tangent to approximate $f(3.02)$. Then calculate the error "exact value minus approximation". (2 pts)

# Solutions EV1-T2

**MCQ:** 1 D, 2 C, 3 B, 4 D, 5 A, 6 A, 7 C, 8 A, 9 B, 10 C, 11 D, 12 C. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

## Exercise 1 – Two growth processes

1. A arithmetic, B geometric; $a_2 = 50, b_2 = 120$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. A: $\dfrac{4(30+60)}{2} = 180$; B: $30(1+2+4+8) = 450$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. $b_1 = 60 > a_1 = 40$. For $n \geq 1$, the increment of B is $30 \times 2^n \geq 60$, greater than 10; each step increases the gap. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 2 – Imposed roots

1. $f = a(x+1)(x-5)$; $-5a = 10$, so $a = -2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. Negative coefficient; f positive between the roots, negative elsewhere. Solutions $(-\infty, -1] \cup [5, +\infty)$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

3. $f = -2x^2 + 8x + 10 = -2(x-2)^2 + 18$. Maximum 18 at 2. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 3 – A controlled approximation

1. Rate $6 + h$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

2. $f\prime(3) = 6$, $f(3) = 10$; $y = 6x - 8$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*

3. Approximation 10.12; exact 10.1204. Exact minus approximate $0.0004 = h^2$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

## EV1-A – Common test

**Paper reserved for the teacher**. Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

### Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $5 + 2 \times 3$.
   A. 14 B. 11 C. 10 D. 21
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^5}{2^3}$.
   A. 2 B. 1 C. 10 D. 4
4. A value of 50 increases by 20%. Final value?
   A. 10 B. 60 C. 70 D. 40
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 100 B. 60 C. 64 D. 96
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{9}{10}$ B. 1 C. $\frac{21}{20}$ D. $\frac{4}{5}$
7. The solution of $2x + 5 = 0$ is
   A. 10 B. $\frac{-5}{2}$ C. $-10$ D. $\frac{5}{2}$
8. The slope between $(0, 5)$ and $(2, 11)$ is
   A. 6 B. 3 C. 2 D. $-3$
9. The mean of $5, 7, 9$ is
   A. 8 B. 21 C. 5 D. 7
10. A sector of $90^\circ$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{2}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$
12. For $f(x) = x^2 + 5$, $f(-2)$ is
    A. 9 B. 7 C. 1 D. 20

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Decreasing reserve (5 points)

A tank initially contains 192 L of water. At each step, half of the water still present is removed, with no water added. Let $u_n$ be the volume, in litres, remaining after $n$ steps; index 0 corresponds to the initial state.

1. Define the sequence with its first term and common ratio. (1 pt)
2. Calculate the volumes for indices 0 to 4 and their sum. (2 pts)
3. Determine the first index where the volume is strictly less than 10 L; describe a threshold loop. (2 pts)

### Exercise 2 – Parameter and tangent (5 points)

For a fixed real number $m$, define on $\mathbb{R}$ the function $f_m(x) = x^2 - 2mx + 4$. The number $m$ is a parameter. The first two questions concern all its values; the last concerns only $m = 1$.

1. Calculate the discriminant of the polynomial $f_m$ in terms of $m$. (1 pt)
2. Determine, according to $m$, the number of real solutions of $f_m(x) = 0$. (2 pts)
3. Now fix $m = 1$. Calculate the rate of change between 2 and $2 + h$, for $h \neq 0$, then the equation of the tangent at the point with x-coordinate 2. (2 pts)

### Exercise 3 – Locus via the midpoint (4 points)

Two fixed points $A$ and $B$ of the plane are 8 units apart. Let $I$ be the midpoint of the segment $[AB]$ and $M$ a variable point of the plane. We seek to describe sets of points starting from the scalar product $\overrightarrow{MA} \cdot \overrightarrow{MB}$.

1. Express $\overrightarrow{MA} \cdot \overrightarrow{MB}$ in terms of $MI$, justifying the identity used. (1 pt)
2. Describe the set of points $M$ for which this product equals 9. (1 pt)
3. Describe in turn the sets obtained when the product equals $-16$, then $-20$. (2 pts)

## Solutions EV1-A

**MCQ:** 1 B, 2 A, 3 D, 4 B, 5 A, 6 B, 7 B, 8 B, 9 D, 10 C, 11 A, 12 A. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

### Exercise 1 – Decreasing reserve

1. $u_0 = 192$, $u_{n+1} = \dfrac{u_n}{2}$; common ratio $\dfrac{1}{2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $192, 96, 48, 24, 12$, sum 372. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $u_4 = 12$, $u_5 = 6$: index 5. Initialise n=0,u=192; while u is at least 10, divide by 2 and increase n. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 2 – Parameter and tangent

1. $\Delta = 4m^2 - 16 = 4(m^2 - 4)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Two roots if $m < -2$ or $m > 2$; one if $m = \pm 2$; none if $-2 < m < 2$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. For m=1, rate $2 + h$, derivative 2 and $f(2) = 4$; tangent $y = 2x$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 3 – Locus via the midpoint

1. $MI^2 - 16$, by scalar identities. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $MI^2 = 25$: circle with centre I, radius 5. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. For -16, $MI^2 = 0$: I alone. For -20, $MI^2 = -4$: empty. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

## EV1-B – Common test

**Paper reserved for the teacher**. Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

### Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $6 + 2 \times 3$.
   A. 11 B. 12 C. 15 D. 24
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{6}$ C. $\frac{2}{5}$ D. $\frac{5}{3}$
3. Calculate $\dfrac{2^6}{2^4}$.
   A. 2 B. 11 C. 1 D. 4
4. A value of 50 increases by 20%. Final value?
   A. 60 B. 70 C. 10 D. 40
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 64 B. 96 C. 60 D. 100
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{21}{20}$ B. $\frac{4}{5}$ C. 1 D. $\frac{9}{10}$
7. The solution of $2x + 6 = 0$ is
   A. 3 B. 12 C. $-3$ D. $-12$
8. The slope between $(0, 6)$ and $(2, 12)$ is
   A. $-3$ B. 3 C. 6 D. 2
9. The mean of $6, 8, 10$ is
   A. 8 B. 24 C. 6 D. 9
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$
12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 8 B. 24 C. 10 D. 2

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Growth by pattern (5 points)

A family of patterns is indexed by the integers $n \geq 1$. The number of dots in the pattern of index $n$ is given by $u_n = n^2 + 2n$. We are interested in particular in the first pattern with at least 99 dots; no figure is needed to use the formula given.

1. Calculate $u_1, u_2, u_3$. (1 pt)
2. Calculate $u_{n+1} - u_n$ then establish the variation. (2 pts)
3. From which index does the pattern contain at least 99 dots? Justify without an endless sweep. (2 pts)

### Exercise 2 – Two dimensions and a constraint (5 points)

A rectangle has a perimeter of 26 m and an area of 40 square metres. Let $x$ and $y$ be its two dimensions, in metres. Both numbers are strictly positive. We will then compare this rectangle with the other rectangles having the same perimeter.

1. Write the relations between the two lengths x and y. (1 pt)
2. Determine the dimensions using a quadratic equation. (2 pts)
3. Among the rectangles with the same perimeter, determine the maximum area and compare it with 40. (2 pts)

### Exercise 3 – Angle then approximation (4 points)

Questions 1 and 2 concern two vectors $\vec{u}$ and $\vec{v}$ with norms 2 and 3 respectively, such that $\vec{u} \cdot \vec{v} = 3$. Question 3 is independent: it concerns approximation by a tangent to the square function.

1. Determine the measure in degrees of the unoriented angle between $\vec{u}$ and $\vec{v}$. (1 pt)
2. Calculate the exact value of $\|\vec{u} - \vec{v}\|$. (1 pt)
3. Independently, the tangent to the square function at 1 has equation $y = 2x - 1$. Use it to approximate $1.03^2$, then calculate the difference between the exact value and the approximation. (2 pts)

# Solutions EV1-B

**MCQ:** 1 B, 2 B, 3 D, 4 A, 5 D, 6 C, 7 C, 8 B, 9 A, 10 C, 11 B, 12 C. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

## Exercise 1 – Growth by pattern

1. $3, 8, 15$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Difference $2n + 3 > 0$: strictly increasing. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $n^2 + 2n - 99 = (n - 9)(n + 11) \geq 0$ with $n \geq 1$: $n \geq 9$; $u_9 = 99$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 2 – Two dimensions and a constraint

1. $x + y = 13$, $xy = 40$, x,y positive. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Roots of $t^2 - 13t + 40 = (t - 5)(t - 8)$: 5 m and 8 m. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $A(x) = x(13 - x) = \frac{169}{4} - (x - \frac{13}{2})^2$, maximum $\frac{169}{4} = 42.25$ m$^2$. Difference $\frac{9}{4}$ m$^2$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 3 – Angle then approximation

1. $\cos\theta = \frac{3}{6} = \frac{1}{2}$, so $60°$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Square $4 + 9 - 6 = 7$; norm $\sqrt{7}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Approximation 1.06; exact 1.0609; error 0.0009. This independent question checks local differentiation. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

# EV2-T1 – Mock exam 1

**Practice paper**. Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14–16 January 2027.

## Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $2 + 2 \times 3$.
   A. 8 B. 12 C. 11 D. 7
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{3}$ C. $\frac{2}{5}$ D. $\frac{5}{6}$
3. Calculate $\dfrac{2^2}{2^0}$.
   A. 2 B. 7 C. 1 D. 4
4. A value of 50 increases by 20%. Final value?
   A. 40 B. 60 C. 70 D. 10
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 96 B. 100 C. 64 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{9}{10}$ C. 1 D. $\frac{21}{20}$
7. The solution of $2x + 2 = 0$ is
   A. $-4$ B. $-1$ C. 1 D. 4
8. The slope between $(0, 2)$ and $(2, 8)$ is
   A. 6 B. 2 C. 3 D. $-3$
9. The mean of $2, 4, 6$ is
   A. 4 B. 2 C. 12 D. 5
10. A sector of $90^\circ$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 4 B. 8 C. 6 D. $-2$

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Manufacturing and quality control (5 points)

Two workshops A and B produce 30 percent and 70 percent respectively of a company's items. Among the items from A, 10 percent are defective; this proportion is 20 percent for B. An item is chosen at random from the total production. The events $A$ and $D$ mean respectively "comes from workshop A" and "is defective".

1. Construct a weighted tree whose first level distinguishes the two workshops. (1 pt)
2. Calculate $P(D)$, then the probability that a defective item comes from A. (2 pts)
3. Are the events $A$ and $D$ independent? Justify numerically. (2 pts)

### Exercise 2 – Minimum of a quotient (5 points)

The function $f$ is defined on the interval $(0, +\infty)$ by $f(x) = x + \dfrac{9}{x}$. We study its variation and certain tangents to its curve. The coordinates requested are to be given in exact form.

1. Calculate and factorise the derivative. (1 pt)
2. Study the variation and give the minimum. (2 pts)
3. Determine the points where the tangent has slope $\dfrac{3}{4}$. (2 pts)

### Exercise 3 – Projection and circle (4 points)

In an orthonormal frame, consider the point $P(5, 2)$ and the line $d$ with equation $x + y - 3 = 0$. Let $H$ be the orthogonal projection of $P$ onto $d$.

1. Give a nonzero normal vector to $d$. (1 pt)
2. Determine the coordinates of $H$ and check the conditions of membership and orthogonality. (1 pt)
3. Give an equation of the circle with centre $P$ passing through $H$. (2 pts)

## Solutions EV2-T1

**MCQ:** 1 A, 2 D, 3 D, 4 B, 5 B, 6 C, 7 B, 8 C, 9 A, 10 C, 11 D, 12 C. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

### Exercise 1 – Manufacturing and quality control

1. First level: 0.3 and 0.7. After A: 0.1 and 0.9; after B: 0.2 and 0.8. Each pair sums to 1. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $P(D) = 0.03 + 0.14 = 0.17$; $P_D(A) = \dfrac{3}{17}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. No: $P_A(D) = 0.1 \neq 0.17$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 2 – Minimum of a quotient

1. $f\prime = \dfrac{(x-3)(x+3)}{x^2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Decreasing up to 3 then increasing; minimum 6. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $1 - \dfrac{9}{x^2} = \dfrac{3}{4}$ gives $x^2 = 36$, so x=6 only on the domain; point $(6, \dfrac{15}{2})$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 3 – Projection and circle

1. $(1, 1)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $\lambda = \dfrac{5+2-3}{2} = 2$; H(3;0). *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. $PH^2 = 8$; $(x-5)^2 + (y-2)^2 = 8$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

## EV2-T2 – Mock exam 1

**Practice paper**. Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14–16 January 2027.

### Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 8 C. 15 D. 9
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{1}{6}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$
3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 2 C. 8 D. 1
4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 60 D. 10
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 64 B. 96 C. 100 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{9}{10}$ B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. 1
7. The solution of $2x + 3 = 0$ is
   A. 6 B. $-6$ C. $\frac{3}{2}$ D. $\frac{-3}{2}$
8. The slope between $(0, 3)$ and $(2, 9)$ is
   A. $-3$ B. 2 C. 6 D. 3
9. The mean of $3, 5, 7$ is
   A. 3 B. 5 C. 6 D. 15
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 12 B. 7 C. $-1$ D. 5

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Two draws (5 points)

An urn contains four red balls and two blue balls, indistinguishable by touch. Two balls are drawn in succession at random, without replacing the first. At each draw, the balls still present are equally likely. The third question proposes another protocol, with replacement.

1. Calculate the probability of drawing two red balls. (1 pt)
2. Calculate the probability of drawing one ball of each colour, taking into account the two possible orders. (2 pts)
3. The first ball is now replaced before the second draw. Calculate the probability of drawing one ball of each colour and compare it with that of the protocol without replacement. (2 pts)

### Exercise 2 – A box (5 points)

Squares of side $x$ cm are cut from the four corners of a square sheet of side 10 cm. Folding up the edges forms a box without a lid. Its volume, in cubic centimetres, is $V(x) = x(10 - 2x)^2$. We study this model on $[0, 5]$, endpoints included even though they are degenerate.

1. Calculate $V'$ in factorised form. (1 pt)
2. Study the variation. (2 pts)
3. Determine the exact maximum volume and justify that the endpoints do not give the maximum. (2 pts)

### Exercise 3 – Recognition and intersection (4 points)

In an orthonormal frame, consider the set $\mathcal{C}$ of points $M(x, y)$ satisfying $x^2 + y^2 + 4x - 2y - 4 = 0$. After identifying it, we will study its intersections with a line and a tangent.

1. Show that $\mathcal{C}$ is a circle and give its centre and radius. (1 pt)
2. Determine all the points common to $\mathcal{C}$ and the line $x = -2$. (1 pt)
3. Write the equation of the tangent to the circle at the intersection point with the greater y-coordinate. Justify its direction. (2 pts)

## Solutions EV2-T2

**MCQ:** 1 D, 2 D, 3 A, 4 C, 5 C, 6 D, 7 D, 8 D, 9 B, 10 A, 11 D, 12 B. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

### Exercise 1 – Two draws

1. $(\frac{4}{6})(\frac{3}{5}) = \frac{2}{5}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $(\frac{4}{6})(\frac{2}{5}) + (\frac{2}{6})(\frac{4}{5}) = \frac{8}{15}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. With replacement: $2(\frac{2}{3})(\frac{1}{3}) = \frac{4}{9}$. The models differ; $\frac{8}{15} > \frac{4}{9}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 2 – A box

1. $V\prime = 4(5-x)(5-3x)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Increasing up to $\frac{5}{3}$ then decreasing up to 5. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $V(\frac{5}{3}) = \frac{2000}{27}$; zero values at the endpoints. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 3 – Recognition and intersection

1. $(x+2)^2 + (y-1)^2 = 9$: centre (-2;1), radius 3. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Points $(-2, 4)$ and $(-2, -2)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Vertical radius at the upper point, horizontal tangent $y = 4$. Normal (0;1). *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

## EV2-A – Mock exam 1

**Paper reserved for the teacher**. Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14–16 January 2027.

### Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $4 + 2 \times 3$.
   A. 13 B. 18 C. 10 D. 9
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{1}{6}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$
3. Calculate $\dfrac{2^4}{2^2}$.
   A. 2 B. 4 C. 1 D. 9
4. A value of 50 increases by 20%. Final value?
   A. 40 B. 10 C. 60 D. 70
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 64 B. 100 C. 96 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{9}{10}$ B. $\frac{4}{5}$ C. 1 D. $\frac{21}{20}$
7. The solution of $2x + 4 = 0$ is
   A. 2 B. 8 C. $-8$ D. $-2$
8. The slope between $(0, 4)$ and $(2, 10)$ is
   A. $-3$ B. 3 C. 2 D. 6
9. The mean of $4, 6, 8$ is
   A. 7 B. 18 C. 6 D. 4
10. A sector of $90^\circ$ represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{1}{5}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$
12. For $f(x) = x^2 + 4$, $f(-2)$ is
    A. 16 B. 0 C. 6 D. 8

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – An unknown probability (5 points)

Two events $A$ and $B$ of the same random experiment satisfy $P(A) = \frac{2}{5}$, $P(B) = \frac{1}{2}$ and $P_A(B) = \frac{3}{4}$. We wish to complete the probabilities of a tree whose first level distinguishes $A$ from its complement.

1. Calculate $P(A \cap B)$. (1 pt)
2. Determine $P_{\overline{A}}(B)$ then complete a tree. (2 pts)
3. Determine $P_B(A)$ and discuss independence. (2 pts)

### Exercise 2 – Relative position and variation (5 points)

The functions $f$ and $g$ are defined on $[-1, 4]$ by $f(x) = x^3 - 3x^2 + 2$ and $g(x) = 2$. Their curves are drawn in the same frame. The variation of $f$ must be distinguished from the position of its curve relative to that of $g$.

1. Calculate $f'$ and study the variation. (1 pt)
2. Compare the curves of f and g on the domain. (2 pts)
3. Determine the minimum and maximum of f on this domain. (2 pts)

### Exercise 3 – A perpendicular bisector (4 points)

In an orthonormal frame, given the points $A(0, 2)$, $B(6, 4)$ and $P(2, 6)$. Let $I$ be the midpoint of $[AB]$. We look for an equation of the perpendicular bisector of this segment, then use this line to compare two distances.

1. Give the midpoint of AB and a normal to its perpendicular bisector. (1 pt)
2. Determine the equation of the perpendicular bisector. (1 pt)
3. Is P equidistant from A and B? Check using two methods. (2 pts)

## Solutions EV2-A

**MCQ:** 1 C, 2 D, 3 B, 4 C, 5 B, 6 C, 7 D, 8 B, 9 C, 10 D, 11 B, 12 D. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

### Exercise 1 – An unknown probability

1. $\frac{3}{10}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $P(\overline{A} \cap B) = \frac{1}{2} - \frac{3}{10} = \frac{1}{5}$; dividing by $\frac{3}{5}$ gives $\frac{1}{3}$; complements $\frac{1}{4}$ and $\frac{2}{3}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $P_B(A) = \frac{3}{5}$; different from $P(A) = \frac{2}{5}$, so not independent. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 2 – Relative position and variation

1. $f' = 3x(x-2)$; increasing up to 0, decreasing up to 2, then increasing. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $f - g = x^2(x-3)$: negative for x<3 except equality at 0, also zero at 3, positive after 3. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $f(-1) = -2, f(0) = 2, f(2) = -2, f(4) = 18$. Minimum -2 at -1 and 2; maximum 18 at 4. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 3 – A perpendicular bisector

1. I(3;3), normal (6;2), or (3;1). *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $3x + y - 12 = 0$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. $3(2) + 6 - 12 = 0$ so yes; $PA^2 = 4 + 16 = 20$, $PB^2 = 16 + 4 = 20$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

# EV2-B – Mock exam 1

**Paper reserved for the teacher**. Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14–16 January 2027.

## Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $5 + 2 \times 3$.
   A. 11 B. 10 C. 14 D. 21
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{1}{6}$ C. $\frac{5}{6}$ D. $\frac{2}{5}$
3. Calculate $\dfrac{2^5}{2^3}$.
   A. 2 B. 10 C. 1 D. 4
4. A value of 50 increases by 20%. Final value?
   A. 10 B. 60 C. 40 D. 70
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 100 B. 64 C. 96 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{9}{10}$ B. 1 C. $\frac{21}{20}$ D. $\frac{4}{5}$
7. The solution of $2x + 5 = 0$ is
   A. $-10$ B. $\frac{5}{2}$ C. $\frac{-5}{2}$ D. 10
8. The slope between $(0, 5)$ and $(2, 11)$ is
   A. 2 B. $-3$ C. 3 D. 6
9. The mean of $5, 7, 9$ is
   A. 7 B. 21 C. 8 D. 5
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 5$, $f(-2)$ is
    A. 1 B. 7 C. 9 D. 20

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Independence and repetition (5 points)

Two events $A$ and $B$ are independent and satisfy $P(A) = \frac{1}{4}$ and $P(B) = \frac{2}{3}$. The first question concerns these events. In a separate experiment, used in questions 2 and 3, a trial with success probability $\frac{1}{4}$ is repeated three times independently.

1. Calculate $P(A \cap B)$ and $P(A \cup B)$. (1 pt)
2. Calculate the probability of exactly two successes in the three trials. (2 pts)
3. Calculate that of at least one success and justify the move to the complement. (2 pts)

### Exercise 2 – Rational function and symmetry (5 points)

The function $f$ is defined on $\mathbb{R} \setminus \{0\}$ by $f(x) = \dfrac{x^2 + 4}{x}$. Its domain is symmetric about $0$, but is not an interval. We will use its parity to link the studies on the two half-lines.

1. Study the parity and rewrite f. (1 pt)
2. Study the variation on $(0, +\infty)$. (2 pts)
3. Deduce the variation on $(-\infty, 0)$ and explain why f is not increasing on its whole domain. (2 pts)

### Exercise 3 – Distance and parameter (4 points)

In an orthonormal frame, given the line $d$ with equation $3x + 4y - 12 = 0$ and the point $P(0, 0)$. Let $H$ be the orthogonal projection of $P$ onto $d$. We look for the circle with centre $P$ tangent to this line.

1. Give a normal and check that $(4, 0)$ is on d. (1 pt)
2. Calculate the projection H of P. (1 pt)
3. Determine the distance from P to d and the equation of the circle with centre P tangent to d. (2 pts)

# Solutions EV2-B

**MCQ:** 1 A, 2 C, 3 D, 4 B, 5 A, 6 B, 7 C, 8 C, 9 A, 10 D, 11 D, 12 C. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

## Exercise 1 – Independence and repetition

1. $\frac{1}{6}$ and $\frac{1}{4}+\frac{2}{3}-\frac{1}{6}=\frac{3}{4}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Three paths, each $(\frac{1}{4})^2(\frac{3}{4})$: total $\frac{9}{64}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Complement of three failures: $1-(\frac{3}{4})^3=\frac{37}{64}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 2 – Rational function and symmetry

1. $f=x+\frac{4}{x}$, odd. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $f'=\frac{x^2-4}{x^2}$; decreasing up to 2 then increasing. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Increasing on $(-\infty,-2]$, decreasing on $[-2,0)$. On $[1,2]$, f decreases: a counterexample to global increase. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 3 – Distance and parameter

1. Normal (3;4); $12-12=0$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $\lambda=-\frac{12}{25}$, so $H(\frac{36}{25},\frac{48}{25})$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Distance $\frac{12}{5}$; $x^2+y^2=\frac{144}{25}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

# EV3-T1 – Mock exam 2

**Practice paper**. Programme studied in chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1–3 April 2027.

## Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $6 + 2 \times 3$.
   A. 12 B. 24 C. 11 D. 15
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{5}{3}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^6}{2^4}$.
   A. 1 B. 11 C. 2 D. 4
4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 70 D. 60
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 64 B. 96 C. 100 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{21}{20}$ B. $\frac{9}{10}$ C. 1 D. $\frac{4}{5}$
7. The solution of $2x + 6 = 0$ is
   A. 12 B. $-12$ C. $-3$ D. 3
8. The slope between $(0, 6)$ and $(2, 12)$ is
   A. $-3$ B. 3 C. 2 D. 6
9. The mean of $6, 8, 10$ is
   A. 9 B. 8 C. 24 D. 6
10. A sector of $90^\circ$ represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{3}{5}$ D. $\frac{1}{5}$
12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 10 B. 8 C. 24 D. 2

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – A compound gain (5 points)

A game consists of two independent trials, each with success probability $\frac{1}{2}$. Each success earns 3 monetary units; taking part costs 2 units. Let $X$ be the net gain, that is, the total received minus the entry fee paid.

1. Establish the probability distribution of the net gain $X$. (1 pt)
2. Calculate its expectation and variance. (2 pts)
3. Determine the entry fee that would make the game fair. Explain what the standard deviation of the net gain would become. (2 pts)

### Exercise 2 – Peak of a model (5 points)

A quantity is modelled for $t \geq 0$ by the function $f(t) = (t+2)e^{-t}$. The time $t$ is expressed in a fixed unit. We look for its maximum value, then an approximation near the initial time.

1. Differentiate f. (1 pt)
2. Study the variation and give the maximum. (2 pts)
3. Write the tangent at 0 and approximate $f(0.1)$. (2 pts)

### Exercise 3 – The unit circle (4 points)

The plane is equipped with an orthonormal frame. On the unit circle with centre $O$, the points $A$ and $B$ are the images of the real numbers $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ respectively.

1. Give their coordinates. (1 pt)
2. Calculate AB. (1 pt)
3. Give the equation of their perpendicular bisector and justify geometrically. (2 pts)

# Solutions EV3-T1

**MCQ:** 1 A, 2 C, 3 D, 4 D, 5 C, 6 C, 7 C, 8 B, 9 B, 10 C, 11 C, 12 A. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

## Exercise 1 – A compound gain

1. Values -2,1,4; probabilities $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $E(X) = 1$; $E(X^2) = 1 + \frac{1}{2} + 4 = \frac{11}{2}$; $V(X) = \frac{9}{2}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Mean gross gain 3: fair fee 3. Standard deviation unchanged, $\frac{3}{\sqrt{2}}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 2 – Peak of a model

1. $f\prime = -(t+1)e^{-t}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Strictly decreasing on $[0, +\infty)$, maximum 2 at 0. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $y = 2 - t$; approximation 1.9. This is not an exact equality. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 3 – The unit circle

1. $A(\frac{\sqrt{3}}{2}, \frac{1}{2})$, $B(-\frac{\sqrt{3}}{2}, \frac{1}{2})$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $AB = \sqrt{3}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Midpoint $(0; \frac{1}{2})$, AB horizontal; vertical perpendicular bisector $x = 0$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

# EV3-T2 – Mock exam 2

**Practice paper**. Programme studied in chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1–3 April 2027.

## Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $2 + 2 \times 3$.
   A. 11 B. 7 C. 12 D. 8
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$
3. Calculate $\dfrac{2^2}{2^0}$.
   A. 7 B. 4 C. 2 D. 1
4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 10 D. 60
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 96 B. 60 C. 100 D. 64
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$
7. The solution of $2x + 2 = 0$ is
   A. $-1$ B. 4 C. $-4$ D. 1
8. The slope between $(0, 2)$ and $(2, 8)$ is
   A. 3 B. 2 C. 6 D. $-3$
9. The mean of $2, 4, 6$ is
   A. 2 B. 4 C. 5 D. 12
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{1}{3}$ D. $\frac{3}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{3}{5}$ C. $\frac{1}{5}$ D. $\frac{2}{5}$
12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 6 B. $-2$ C. 4 D. 8

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Simulation and mean (5 points)

The random variable $X$ takes the values 0, 2 and 6 with respective probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$. We then simulate samples of 25 independent realisations of this variable and let $m$ be the mean observed in a sample.

1. Calculate the expectation $E(X)$. (1 pt)
2. Calculate $V(X)$ then the exact standard deviation $\sigma(X)$. (2 pts)
3. Write the interval of values of $m$ satisfying $|m - E(X)| \leq \dfrac{2\sigma(X)}{5}$. Explain why a simulation on a finite number of samples does not guarantee an exact coverage percentage. (2 pts)

### Exercise 2 – Comparing two models (5 points)

Define on $\mathbb{R}$ the functions $f(x) = e^{2x}$ and $g(x) = 1 + 2x$, then their difference $h = f - g$. The last question considers the values of $f$ at integer times, starting from index 0.

1. Calculate the derivative of $h = f - g$. (1 pt)
2. Study h and compare the curves. (2 pts)
3. For $u_n = e^{2n}$, give the common ratio and the sum of the first four terms. (2 pts)

### Exercise 3 – A displaced circle (4 points)

In an orthonormal frame, consider the circle $\mathcal{C}$ with equation $(x-1)^2 + (y+1)^2 = 4$ and the line $d$ with equation $y = -1$. Call $A$ and $B$ their points of intersection, $A$ being the one with the smaller x-coordinate.

1. Give the centre and radius. (1 pt)
2. Determine the intersections with the line. (1 pt)
3. For M on the circle, calculate $\overrightarrow{MA} \cdot \overrightarrow{MB}$ where A and B are these intersections; justify. (2 pts)

# Solutions EV3-T2

**MCQ:** 1 D, 2 D, 3 B, 4 D, 5 C, 6 C, 7 A, 8 A, 9 B, 10 A, 11 B, 12 A. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

## Exercise 1 – Simulation and mean

1. $E(X) = 0 + \frac{1}{2} + \frac{3}{2} = 2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $E(X^2) = 1 + 9 = 10$, variance 6, standard deviation $\sqrt{6}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $[2 - \frac{2\sqrt{6}}{5}, 2 + \frac{2\sqrt{6}}{5}]$. A frequency over a finite number of simulations fluctuates. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 2 – Comparing two models

1. $h\prime = 2(e^{2x} - 1)$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Decreasing up to 0 then increasing; minimum 0. So $f \geq g$ with equality only at 0. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Common ratio $e^2$; sum $1 + e^2 + e^4 + e^6 = \frac{1 - e^8}{1 - e^2}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 3 – A displaced circle

1. Centre (1;-1), radius 2. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. A(-1;-1), B(3;-1). *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. AB is a diameter, so the product is 0, endpoints included; identity $MI^2 - 4 = 0$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

## EV3-A – Mock exam 2

**Paper reserved for the teacher**. Programme studied in chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1–3 April 2027.

### Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $3 + 2 \times 3$.
   A. 9 B. 15 C. 8 D. 12
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{5}{6}$ C. $\frac{1}{6}$ D. $\frac{5}{3}$
3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 1 C. 8 D. 2
4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 60 D. 70
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 96 B. 60 C. 100 D. 64
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. $\frac{9}{10}$ D. 1
7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. $-6$ D. 6
8. The slope between $(0, 3)$ and $(2, 9)$ is
   A. 6 B. $-3$ C. 2 D. 3
9. The mean of $3, 5, 7$ is
   A. 6 B. 15 C. 3 D. 5
10. A sector of $90^\circ$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{4}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{5}{2}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$
12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 12 B. 7 C. 5 D. $-1$

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Lottery and dispersion (5 points)

A lottery pays a gross gain $G$ of 0, 4 or 12 monetary units with respective probabilities $\frac{1}{2}$, $\frac{3}{8}$ and $\frac{1}{8}$. To take part, one pays 3 units. Let $X = G - 3$ be the net gain.

1. Give the distribution of the net gain X. (1 pt)
2. Calculate the expectation and variance of X. (2 pts)
3. The gross gains and the entry fee are doubled. What become the mean and standard deviation of the net gain? (2 pts)

### Exercise 2 – Derivative and horizontal tangent (5 points)

The function $f$ is defined on $\mathbb{R}$ by $f(x) = (2x - 3)e^x$. We study its extremum and the tangent to its curve at the point with x-coordinate 0. All values are expected in exact form.

1. Calculate $f(0)$, then $f\prime(x)$ and $f\prime(0)$. (1 pt)
2. Study the variation of $f$ on $\mathbb{R}$ and give its exact extremum. (2 pts)
3. Determine the equation of the tangent at the point with x-coordinate 0. State whether its point of contact is above or below the x-axis. (2 pts)

### Exercise 3 – Angle and length (4 points)

In an orthonormal frame, given the points $A(0,0)$, $B(4,0)$ and $C(2,2\sqrt{3})$. We look for the nature of the triangle $ABC$ by comparing lengths and scalar product.

1. Calculate AB and AC. (1 pt)
2. Determine the cosine then the angle BAC in radians. (1 pt)
3. Determine BC and the nature of the triangle. (2 pts)

## Solutions EV3-A

**MCQ:** 1 A, 2 B, 3 A, 4 C, 5 C, 6 D, 7 B, 8 D, 9 D, 10 B, 11 A, 12 B. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

### Exercise 1 – Lottery and dispersion

1. X is -3,1,9 with the same probabilities. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $E(X) = -\frac{3}{2} + \frac{3}{8} + \frac{9}{8} = 0$. $E(X^2) = \frac{9}{2} + \frac{3}{8} + \frac{81}{8} = 15$; variance 15. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Net gain 2X, expectation 0; standard deviation $2\sqrt{15}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 2 – Derivative and horizontal tangent

1. $f(0) = -3$; $f\prime = (2x-1)e^x$, $f\prime(0) = -1$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Decreasing up to $\frac{1}{2}$, increasing after; minimum $-2e^{\frac{1}{2}}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $y = -x - 3$; contact (0;-3), below the horizontal axis. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 3 – Angle and length

1. AB=4, $AC = \sqrt{4+12} = 4$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Product 8; cosine $\frac{8}{16} = \frac{1}{2}$, angle $\frac{\pi}{3}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. $BC = \sqrt{4+12} = 4$: equilateral triangle. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

# EV3-B – Mock exam 2

**Paper reserved for the teacher**. Programme studied in chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1–3 April 2027.

## Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $4 + 2 \times 3$.
   A. 10 B. 9 C. 18 D. 13
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{6}$ C. $\frac{5}{3}$ D. $\frac{2}{5}$
3. Calculate $\dfrac{2^4}{2^2}$.
   A. 4 B. 1 C. 2 D. 9
4. A value of 50 increases by 20%. Final value?
   A. 70 B. 60 C. 10 D. 40
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 96 B. 60 C. 64 D. 100
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{9}{10}$ B. 1 C. $\frac{21}{20}$ D. $\frac{4}{5}$
7. The solution of $2x + 4 = 0$ is
   A. 2 B. 8 C. $-2$ D. $-8$
8. The slope between $(0, 4)$ and $(2, 10)$ is
   A. 2 B. $-3$ C. 3 D. 6
9. The mean of $4, 6, 8$ is
   A. 7 B. 18 C. 6 D. 4
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{1}{5}$ B. $\frac{5}{2}$ C. $\frac{2}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 4$, $f(-2)$ is
    A. 16 B. 0 C. 6 D. 8

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Four trials and a gain (5 points)

A game consists of four independent trials; each trial succeeds with probability $\frac{1}{2}$. The gross gain $G$ is 8 monetary units if all four trials succeed, and 0 otherwise. A stake may be set to make the game fair.

1. Calculate the probability of receiving 8. (1 pt)
2. Calculate the expectation and variance of G. (2 pts)
3. Determine the fair stake; explain whether the expectation is a possible gain. (2 pts)

### Exercise 2 – A derivative of constant sign (5 points)

Consider the function $f$ defined on $\mathbb{R}$ by $f(x) = (x^2 + 1)e^x$. We want to check whether a horizontal tangent necessarily corresponds to an extremum.

1. Calculate and factorise $f'$. (1 pt)
2. Study the variation; does the horizontal tangent at -1 give an extremum? (2 pts)
3. Write the tangent at 0 then calculate $f(1)$ exactly. (2 pts)

### Exercise 3 – Projection onto a diagonal (4 points)

In an orthonormal frame with origin $O$, consider $P(2, 0)$ and the line $d$ with equation $y = x$. Let $H$ be the orthogonal projection of $P$ onto $d$. The unit vector asked for at the end must be oriented from $O$ towards $H$.

1. Give a nonzero normal vector to the line $d$. (1 pt)
2. Determine the coordinates of $H$, then the distance $PH$. (1 pt)
3. Give the unoriented angle $\widehat{POH}$ in radians, then the coordinates of the vector $\dfrac{\overrightarrow{OH}}{OH}$. (2 pts)

# Solutions EV3-B

**MCQ:** 1 A, 2 B, 3 A, 4 B, 5 D, 6 B, 7 C, 8 C, 9 C, 10 D, 11 D, 12 D. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

## Exercise 1 – Four trials and a gain

1. $(\frac{1}{2})^4 = \frac{1}{16}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $E(G) = \frac{1}{2}$, $E(G^2) = 4$; $V(G) = \frac{15}{4}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Stake $\frac{1}{2}$. The expectation of the gross gain is $\frac{1}{2}$, not possible in a single game. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 2 – A derivative of constant sign

1. $f\prime = (x^2 + 2x + 1)e^x = (x+1)^2 e^x$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Positive except at -1; f is strictly increasing (on both sides and by continuity). No extremum at -1. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $f(0) = 1$, $f\prime(0) = 1$: $y = x + 1$; $f(1) = 2e$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 3 – Projection onto a diagonal

1. Normal (1;-1). *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. H(1;1), PH=$\sqrt{2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Angle $\frac{\pi}{4}$; vector $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

# EV4-T1 – EAM preparation

**Practice paper**. Whole programme.

Duration: 2 h. No calculator. Marked out of 20. Practice sessions in May and June; EAM date for the zone to be confirmed.

## Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $5 + 2 \times 3$.
   A. 14 B. 11 C. 21 D. 10
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{5}{6}$ C. $\frac{1}{6}$ D. $\frac{2}{5}$
3. Calculate $\dfrac{2^5}{2^3}$.
   A. 1 B. 4 C. 10 D. 2
4. A value of 50 increases by 20%. Final value?
   A. 40 B. 10 C. 60 D. 70
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 60 B. 96 C. 100 D. 64
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. 1 C. $\frac{9}{10}$ D. $\frac{21}{20}$
7. The solution of $2x + 5 = 0$ is
   A. $\frac{5}{2}$ B. 10 C. $\frac{-5}{2}$ D. $-10$
8. The slope between $(0, 5)$ and $(2, 11)$ is
   A. 3 B. $-3$ C. 6 D. 2
9. The mean of $5, 7, 9$ is
   A. 7 B. 21 C. 5 D. 8
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$
12. For $f(x) = x^2 + 5$, $f(-2)$ is
    A. 20 B. 7 C. 1 D. 9

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Choosing a distribution (5 points)

We compare two games. The first gives a gain $X = 2$ with certainty. The second gives a gain $Y = -1$ with probability $\frac{2}{3}$ and a gain $Y = 8$ with probability $\frac{1}{3}$. The gains are expressed in the same monetary unit; a negative value represents a loss.

1. Calculate $E(X)$ and $E(Y)$. (1 pt)
2. Calculate $V(Y)$ and compare the dispersions of the two gains. (2 pts)
3. Both games are played and the gains added. Calculate $E(X+Y)$, citing the property used. (2 pts)

### Exercise 2 – Discrete threshold and continuous model (5 points)

A quantity is modelled by $Q(t) = 64e^{-t}$ for $t \geq 0$. It is observed at integer times $n \in \mathbb{N}$ and we set $u_n = Q(n)$. For comparisons without a calculator, the bounds $e^2 < 8 < e^3$ are given.

1. Study the direction of variation. (1 pt)
2. Determine the first integer n such that $Q(n) < 8$ using the bounds provided. (2 pts)
3. Give the type of $u_n = Q(n)$ and the expression of $u_0 + \cdots + u_3$. (2 pts)

### Exercise 3 – Equation and circle (4 points)

In an orthonormal frame, consider the set $\mathcal{C}$ defined by $x^2 + y^2 - 2x - 4y + 1 = 0$. The task is to determine its characteristics, then to identify a particular tangent.

1. Show that $\mathcal{C}$ is a circle and state its centre and radius. (1 pt)
2. Determine its points of intersection with the line with equation $x = 1$. (1 pt)
3. Determine the equation of the tangent to the circle at the point with the greatest y-coordinate and justify. (2 pts)

## Solutions EV4-T1

**MCQ:** 1 B, 2 B, 3 B, 4 C, 5 C, 6 B, 7 C, 8 A, 9 A, 10 A, 11 A, 12 D. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

### Exercise 1 – Choosing a distribution

1. $E(X) = 2$; $E(Y) = -\frac{2}{3} + \frac{8}{3} = 2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $E(Y^2) = \frac{2}{3} + \frac{64}{3} = 22$, variance 18, against 0 for X. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Linearity: expectation 4. X is also constant. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 2 – Discrete threshold and continuous model

1. $Q\prime = -64e^{-t} < 0$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $Q(2) > 8$, $Q(3) < 8$; decreasing: first index 3. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Geometric, first term 64, common ratio $e^{-1}$; sum $\frac{64(1-e^{-4})}{1-e^{-1}}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 3 – Equation and circle

1. $(x-1)^2 + (y-2)^2 = 4$; centre (1;2), radius 2. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. (1;0) and (1;4). *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Vertical radius at the point (1;4), horizontal tangent $y = 4$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

# EV4-T2 – EAM preparation

**Practice paper**. Whole programme.

Duration: 2 h. No calculator. Marked out of 20. Practice sessions in May and June; EAM date for the zone to be confirmed.

## Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $6 + 2 \times 3$.
   A. 24 B. 15 C. 11 D. 12
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^6}{2^4}$.
   A. 2 B. 4 C. 1 D. 11
4. A value of 50 increases by 20%. Final value?
   A. 60 B. 10 C. 70 D. 40
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 64 B. 100 C. 96 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$
7. The solution of $2x + 6 = 0$ is
   A. $-3$ B. 12 C. 3 D. $-12$
8. The slope between $(0, 6)$ and $(2, 12)$ is
   A. 2 B. $-3$ C. 3 D. 6
9. The mean of $6, 8, 10$ is
   A. 6 B. 24 C. 8 D. 9
10. A sector of $90^\circ$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 10 B. 8 C. 24 D. 2

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – A reversed tree (5 points)

Two events $A$ and $B$ satisfy $P(A) = \dfrac{3}{5}$, $P_A(B) = \dfrac{1}{3}$ and $P_{\overline{A}}(B) = \dfrac{3}{4}$. We wish to build a tree that starts by distinguishing $B$ and $\overline{B}$, instead of $A$ and $\overline{A}$.

1. Calculate $P(A \cap B)$ and $P(B)$. (1 pt)
2. Calculate $P_B(A)$ and $P_{\overline{B}}(A)$. (2 pts)
3. Construct and complete the weighted tree whose first level distinguishes $B$ and $\overline{B}$, then the second $A$ and $\overline{A}$. (2 pts)

### Exercise 2 – Two optimisation methods (5 points)

The function $f$ is defined on the interval $[1, 6]$ by $f(x) = x(8 - x)$. We look for its extrema on this precise domain, then for the values of $x$ for which $f(x)$ reaches at least 15.

1. Give a vertex form and a maximum. (1 pt)
2. Recover the maximum by differentiation. (2 pts)
3. Determine the minimum on the domain and solve $f(x) \geq 15$. (2 pts)

### Exercise 3 – Arguments and symmetry (4 points)

In an orthonormal frame with origin $O$, the points $A$ and $B$ of the unit circle are the images of $\dfrac{7\pi}{6}$ and $\dfrac{11\pi}{6}$. The unoriented angle $\widehat{AOB}$ will be given in radians, between 0 and $\pi$.

1. Give their exact coordinates. (1 pt)
2. Calculate $\overrightarrow{OA} \cdot \overrightarrow{OB}$. (1 pt)
3. Determine the unoriented angle AOB then AB. (2 pts)

# Solutions EV4-T2

**MCQ:** 1 D, 2 A, 3 B, 4 A, 5 B, 6 C, 7 A, 8 C, 9 C, 10 D, 11 D, 12 A. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

## Exercise 1 – A reversed tree

1. $P(A \cap B) = \frac{1}{5}$; $P(B) = \frac{1}{5} + \frac{3}{10} = \frac{1}{2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $P_B(A) = \frac{2}{5}$; $P(A \cap \overline{B}) = \frac{2}{5}$ so $P_{\overline{B}}(A) = \frac{4}{5}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. First level $\frac{1}{2}$, $\frac{1}{2}$; after B: $\frac{2}{5}$ and $\frac{3}{5}$; after the complement of B: $\frac{4}{5}$ and $\frac{1}{5}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 2 – Two optimisation methods

1. $f = 16 - (x-4)^2$; maximum 16 at 4. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $f\prime = 8 - 2x$, positive before 4 and negative after. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Endpoints: f(1)=7, f(6)=12; minimum 7. Inequality $(x-4)^2 \leq 1$: $[3,5]$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 3 – Arguments and symmetry

1. $A(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$, $B(\frac{\sqrt{3}}{2}, -\frac{1}{2})$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $-\frac{3}{4} + \frac{1}{4} = -\frac{1}{2}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Angle $\frac{2\pi}{3}$; $AB = \sqrt{3}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

## EV4-A – EAM preparation

**Paper reserved for the teacher**. Whole programme.

Duration: 2 h. No calculator. Marked out of 20. Practice sessions in May and June; EAM date for the zone to be confirmed.

### Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $2 + 2 \times 3$.
   A. 11 B. 7 C. 8 D. 12
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{1}{6}$ C. $\frac{5}{6}$ D. $\frac{2}{5}$
3. Calculate $\dfrac{2^2}{2^0}$.
   A. 2 B. 1 C. 7 D. 4
4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 60 D. 10
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 100 B. 60 C. 96 D. 64
6. A 25% increase then a 20% decrease: overall coefficient?
   A. 1 B. $\frac{4}{5}$ C. $\frac{21}{20}$ D. $\frac{9}{10}$
7. The solution of $2x + 2 = 0$ is
   A. 1 B. 4 C. $-1$ D. $-4$
8. The slope between $(0, 2)$ and $(2, 8)$ is
   A. $-3$ B. 3 C. 6 D. 2
9. The mean of $2, 4, 6$ is
   A. 4 B. 5 C. 2 D. 12
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$
12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 6 B. $-2$ C. 4 D. 8

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – Two linked variables (5 points)

The random variable $X$ takes the values 1, 2 and 3, each with probability $\frac{1}{3}$. On the same experiment, define the variable $Y = 4 - X$. We wish to check whether a rule valid for expectation remains true for variance.

1. Calculate $E(X)$ and $E(Y)$. (1 pt)
2. Calculate $V(X)$, $V(Y)$ and $V(X+Y)$. (2 pts)
3. Use these results to refute the statement: "The variance of a sum is always equal to the sum of the variances." (2 pts)

### Exercise 2 – Comparing a curve and a line (5 points)

The functions $f(x) = e^{-x}$ and $g(x) = 1 - x$ are defined on $\mathbb{R}$. Let $h = f - g$. Comparing their curves will then allow us to check the direction of an approximation.

1. Differentiate their difference. (1 pt)
2. Prove $f(x) \geq g(x)$ for every real number x. (2 pts)
3. Write the affine approximation of $e^{-0.02}$ and state whether it is from below or from above. (2 pts)

### Exercise 3 – Normal vector and radius (4 points)

In an orthonormal frame, consider the point $I(1,1)$ and the line $d$ with equation $3x + 4y - 17 = 0$. Let $H$ be the orthogonal projection of $I$ onto $d$. We look for the circle with centre $I$ tangent to $d$.

1. Calculate the distance from I to d. (1 pt)
2. Calculate the projection H. (1 pt)
3. Give the equation of the circle with centre I tangent to d and check that H belongs to the circle. (2 pts)

## Solutions EV4-A

**MCQ:** 1 C, 2 C, 3 D, 4 C, 5 A, 6 A, 7 C, 8 B, 9 A, 10 A, 11 B, 12 A. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

### Exercise 1 – Two linked variables

1. Both expectations equal 2. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $E(X^2) = \frac{14}{3}$, so $V(X) = \frac{2}{3}$; same variance for Y. X+Y=4, variance 0. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $0 \neq \frac{2}{3} + \frac{2}{3}$. The counterexample is on a single sample space; Y depends on X. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 2 – Comparing a curve and a line

1. $h\prime = 1 - e^{-x}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Negative if x<0, positive if x>0; minimum h(0)=0. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Approximation 0.98; from below, thanks to the inequality. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

### Exercise 3 – Normal vector and radius

1. $\frac{|3+4-17|}{5} = 2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $\lambda = -\frac{10}{25} = -\frac{2}{5}$; H$(\frac{11}{5};\frac{13}{5})$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. $(x-1)^2 + (y-1)^2 = 4$; for H, $(\frac{6}{5})^2 + (\frac{8}{5})^2 = 4$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.

## EV4-B – EAM preparation

**Paper reserved for the teacher**. Whole programme.

Duration: 2 h. No calculator. Marked out of 20. Practice sessions in May and June; EAM date for the zone to be confirmed.

### Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $3 + 2 \times 3$.
   A. 9 B. 12 C. 15 D. 8
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{5}{6}$ C. $\frac{2}{5}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^3}{2^1}$.
   A. 8 B. 2 C. 4 D. 1
4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 60 D. 10
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 60 B. 64 C. 96 D. 100
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{21}{20}$ B. 1 C. $\frac{4}{5}$ D. $\frac{9}{10}$
7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. 6 C. $\frac{-3}{2}$ D. $-6$
8. The slope between $(0, 3)$ and $(2, 9)$ is
   A. 6 B. $-3$ C. 3 D. 2
9. The mean of $3, 5, 7$ is
   A. 3 B. 15 C. 6 D. 5
10. A sector of $90^\circ$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$
12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 12 B. 5 C. $-1$ D. 7

## Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

### Exercise 1 – An average price (5 points)

A service offers several rates. 60 percent of customers choose plan A; among them, half pay 0 units and the other half pay 4 units. All the other customers pay 6 units. A customer is chosen at random and $X$ is the price they pay. The last question studies a uniform reduction, including for the rate that was initially zero.

1. Establish the probability distribution of $X$. (1 pt)
2. Calculate its expectation and variance. (2 pts)
3. All prices fall by 1 unit. Calculate the new mean and the new standard deviation. Interpret the negative price obtained for some customers. (2 pts)

### Exercise 2 – Exponential maximum (5 points)

The function $f$ is defined on $[0,4]$ by $f(t) = t^2 e^{-t}$. We look for its maximum on this interval. For the last comparison without a calculator, you may use the inequality given, $e^2 > 4$.

1. Calculate $f'$ in factorised form. (1 pt)
2. Determine the variation and the maximum. (2 pts)
3. Compare $f(4)$ exactly with the maximum, given $e^2 > 4$. (2 pts)

### Exercise 3 – Determining a position (4 points)

In an orthonormal frame, the points $A(0,0)$ and $B(4,0)$ are fixed. The point $M$ moves on the line with equation $y = 2$; its coordinates are therefore $M(x,2)$, where $x$ ranges over $\mathbb{R}$.

1. Calculate $\overrightarrow{MA} \cdot \overrightarrow{MB}$. (1 pt)
2. Determine the values of x for which the two vectors are orthogonal. (1 pt)
3. Interpret the result using the circle with diameter AB. (2 pts)

# Solutions EV4-B

**MCQ:** 1 A, 2 B, 3 C, 4 C, 5 D, 6 B, 7 C, 8 C, 9 D, 10 C, 11 A, 12 D. Each correct answer: 0.5 point; total 6.

**Checks of the fluency questions.** Order of operations; sum of fractions; subtraction of exponents; multiplying coefficient 1.2; division by 0.8; product 1.25 times 0.8; solving an affine equation; vertical increment divided by horizontal; mean; angle divided by 360; complement to 1; substitution in brackets.

## Exercise 1 – An average price

1. Values 0,4,6; probabilities 0.3; 0.3; 0.4. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. $E(X) = 3.6 = \frac{18}{5}$; $E(X^2) = 19.2 = \frac{96}{5}$; variance $\frac{156}{25}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. Mean $\frac{13}{5}$; standard deviation $\frac{2\sqrt{39}}{5}$, unchanged. The price -1 would be a refund: check whether this policy is really intended. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 2 – Exponential maximum

1. $f\prime = t(2-t)e^{-t}$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Increasing from 0 to 2 then decreasing; maximum $4e^{-2}$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*
3. $f(4) = 16e^{-4}$; ratio to the maximum $4e^{-2} < 1$ since $e^2 > 4$. *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

## Exercise 3 – Determining a position

1. $(-x,-2)\cdot(4-x,-2) = x^2 - 4x + 4 = (x-2)^2$. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
2. Product zero only at x=2. *Marking: 1 point. Result and basic justification expected; half a point possible for a correct but unfinished approach.*
3. Circle with centre (2;0), radius 2; the line y=2 is tangent to it at (2;2). *Marking: 2 points. Method or justification: 1; calculation and conclusion: 1.*

**Calibration.** Target duration: 20–25 min for the fluency questions, 80–85 min for the exercises, 10–15 min for proofreading. These durations are editorial estimates to be measured in class. Accept any correct method; do not penalise twice a numerical error that is propagated consistently. The proposed marking scheme is not a decision of the team to harmonise marking.